PHYSICS

ROTATIONAL DYNAMICS_MICRO_XI



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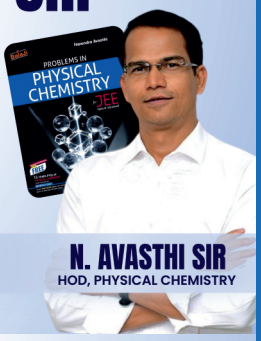

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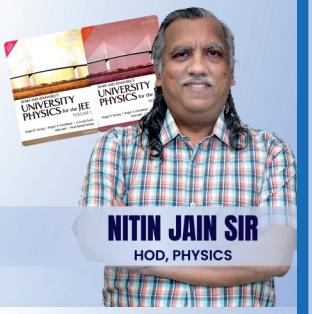


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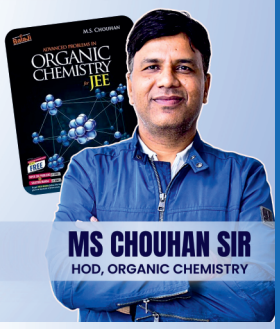


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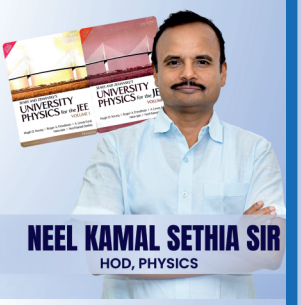


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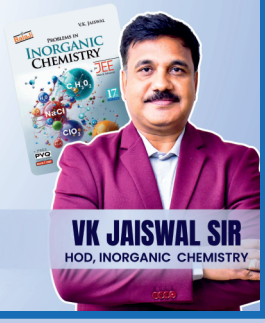


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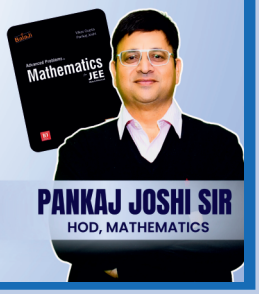


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MICRO-XI

ROTATIONAL DYNAMICS

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EXERCISE-I

Only one option is correct

1. Three bodies have equal masses m. Body A is solid cylinder of radius R, body B is a square lamina of side R, and body C is a solid sphere of radius R. Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)

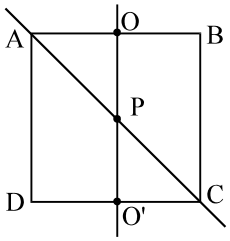
(A) A (B) B (C) C (D) A and C both

2. For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body

(A) a disc of radius a (B) a ring of radius a

(C) a square lamina of side 2a (D) four rods forming a square of side 2a

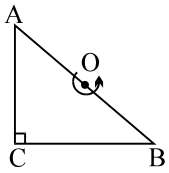
3. Let I_1 and I_2 be the moment of inertia of a uniform square plate about axes APC and OPO' respectively as shown in the figure. P is centre of square. The ratio $\frac{I_1}{I_2}$ of moment of inertia is



(A) $\frac{1}{\sqrt{2}}$ (B) 2

(C) $\frac{1}{2}$ (D) 1

4. Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M, side AC = BC = a about an axis perpendicular to the plane of the plate and passing through the mid point of side AB

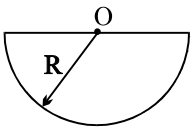


(A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$ (C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$

5. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bend in the form of a semicircular arc. Now its moment of inertia through the centre of the semi circular arc and perpendicular to its plane is I_2. The ratio of $I_1 : I_2$ will be _____

(A) < 1 (B) > 1 (C) = 1 (D) can't be said

6. Moment of inertia of a thin semicircular disc (mass = M & radius = R) about an axis through point O and perpendicular to plane of disc, is given by :



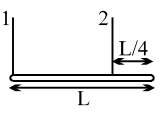
(A) $\frac{1}{4}$ MR² (B) $\frac{1}{2}$ MR² (C) $\frac{1}{8}$ MR² (D) MR²

7. A weightless rod is acted on by upward parallel forces of 2N and 4N ends A and B respectively. The total length of the rod AB = 3m. To keep the rod in equilibrium a force of 6N should act in the following manner:

(A) Downwards at any point between A and B. (B) Downwards at mid point of AB.

(C) Downwards at a point C such that AC = 1m. (D) Downwards at a point D such that BD = 1m.

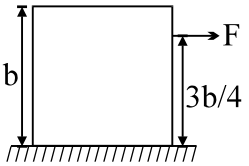
8. A heavy rod of length L and weight W is suspended horizontally by two vertical ropes as shown. The first rope is attached to the left end of the rod while the second rope is attached a distance L/4 from right end. The tension in the second rope is:



(A) (1/2)W (B) (1/4)W (C) (1/3)W (D) (2/3)W

(E) W

9. A uniform cube of side 'b' and mass M rest on a rough horizontal table. A horizontal force F is applied normal to one of the face at a point, at a height 3b/4 above the base. What should be the coefficient of friction (μ) between cube and table so that is will tip about an edge before it starts slipping?



(A) $\mu > \frac{2}{3}$ (B) $\mu > \frac{1}{3}$

(C) $\mu > \frac{3}{2}$ (D) none

10. A heavy seesaw (i.e., not massless) is out of balance. A light girl sits on the end that is tilted downward, and a heavy body sits on the other side so that the seesaw now balances. If they both move forward so that they are one-half their original distance from the pivot point (the fulcrum) what will happen to the seesaw ?

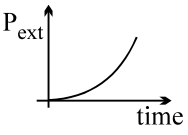
(A) The side the body is sitting on will tilt downward

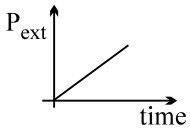
(B) The side the girl is sitting on will once again tilt downward

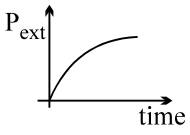
(C) Nothing ; the seesaw will still be balanced

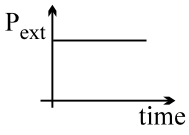
(D) It is impossible to say without knowing the masses and the distances

11. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :

(A) 

(B) 

(C) 

(D) 

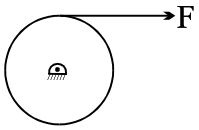
12. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As the time increases,

(A) its angular velocity increases, but force on hinge remains constant

(B) its angular velocity remains same, but force on hinge increases

(C) its angular velocity increases and force on hinge increases

(D) its angular velocity remains same and force on hinge is constant

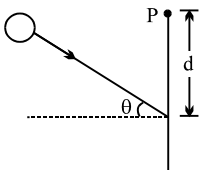


13. A man, sitting firmly over a rotating stool has his arms streched. If he folds his arms, the work done by the man is

(A) zero (B) positive

(C) negative (D) may be positive or negative.

14. A ball of mass m moving with velocity v, collide with the wall elastically as shown in the figure. After impact the change in angular momentum about P is:



(A) 2 *mvd* (B) 2 *mvd* cosθ

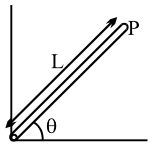
(C) 2 *mvd* sin θ (D) zero

15. Two uniform spheres of mass M have radii R and 2R. Each sphere is rotating about a fixed axis through a diameter. the rotational kinetic energies of the spheres are identical. What is the ratio of the magnitude of the angular momenta of these spheres? That is, $\frac{L_{2R}}{L_R}$ =

(A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$ (E) 1

16. A spinning ice skater can increase his rate of rotation by bringing his arms and free leg closer to his body. How does this procedure affect the skater's angular momentum and kinetic energy?
(A) angular momentum remains the same while kinetic energy increases
(B) angular momentum remains the same while kinetic energy decreases
(C) both angular momentum and kinetic energy remain the same
(D) angular momentum increases while kinetic energy remains the same
(E) both angular momentum and kinetic energy increase

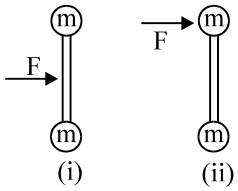
17. A uniform flag pole of length L and mass M is pivoted on the ground with a frictionless hinge. The flag pole makes an angle θ with the horizontal. The moment of inertia of the flag pole about one end is $(1/3)ML^2$. If it starts falling from the position shown in the accompanying figure, the linear acceleration of the free end of the flag pole — labeled P – would be:
(A) $(2/3)\,g\cos\theta$
(B) $(2/3)\,g$
(C) g
(D) $(3/2)\,g\cos\theta$
(E) $(3/2)\,g$



18. A mass *m* is moving at speed *v* perpendicular to a rod of length *d* and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is
(A) $2v/d$ (B) $2v/(3d)$ (C) v/d (D) $3v/(2d)$

19. The moment of inertia of a solid cylinder about its axis is given by $(1/2)MR^2$. If this cylinder rolls without slipping, the ratio of its rotational kinetic energy to its translational kinetic energy is
(A) 1 : 1 (B) 2 : 2 (C) 1 : 2 (D) 1 : 3

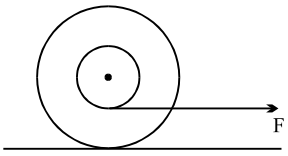
20. A force F is applied to a dumbbell for a time interval, t, first as in (i) and then as in (ii).
In which case does the dumbbell acquire the greater centre-of-mass speed?
(A) (i)
(B) (ii)
(C) there is no difference
(D) the answer depends on the rotational inertia of the dumbbell



21. A hoop and a solid cylinder have the same mass and radius. They both roll, without slipping, on a horizontal surface. If their kinetic energies are equal
(A) the hoop has a greater translational speed than the cylinder
(B) the cylinder has a greater translational speed than the hoop
(C) the hoop and the cylinder have the same translational speed
(D) the hoop has a greater rotational speed than the cylinder

22. A wheel of radius r rolling on a straight line, the velocity of its centre being v. At a certain instant the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The incorrect statement is:
(A) The velocity of any point P of the wheel is proportional to MP.
(B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v.
(C) The point of contact M is instantaneously at rest.
(D) The velocities of any two parts of the wheel which are equidistant from centre are equal.

23. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in fig. then in case of pure rolling



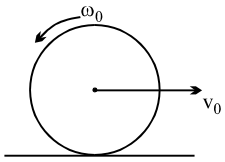
(A) Thread unwinds, spool rotates anticlockwise and friction act leftwards
(B) Thread winds, spool rotates clockwise and friction acts leftwards
(C) Thread winds, spool moves to the right and friction act rightwards
(D) Thread winds, spool moves to the right and friction does not come into existence

24. A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :
(A) thin pipe (B) uniform cylinder
(C) uniform sphere (D) thin spherical shell

25. A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere:
(A) must be forward (in direction of v) (B) must be backward (opposite to v)
(C) cannot be zero (D) none of the above

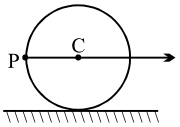
26. A cylinder is pure rolling up an incline plane. It stops momentarily and then rolls back. The force of friction
(A) on the cyclinder is zero throughout the journey
(B) is directed opposite to the velocity of the centre of mass throughout the journey
(C) is directed up the plane throughout the journey
(D) is directed down the plane throughout the journey

27. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



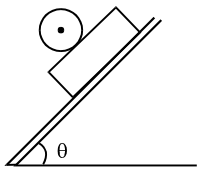
(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

28. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is
(A) zero
(B) 45°
(C) 135°
(D) $\tan^{-1}(1/2)$



29. A solid uniform disk of mass m rolls without slipping down a fixed inclined plane with an acceleration a. The frictional force on the disk due to surface of the plane is :
(A) 2 ma (B) 3/2 ma
(C) ma (D) 1/2 ma

30. A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is
(A) up the plane
(B) down the plane
(C) horizontal
(D) zero



31. When the speed of a front wheel drive car is increasing on a horizontal road the direction of the frictional force on the tires is :

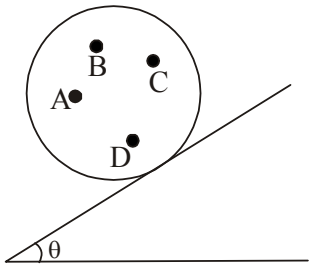
(A) forward for all tires

(B) backward for all tires

(C) forward for the front tires and backward for the rear tires

(D) backward for the front tires and forward for the rear tires

32. A non uniform sphere can be kept on a rough inclined plane so that it is in equilibrium. In the figure below the dots represents location of centre of mass. In which one of the positions can sphere be in equilibrium?



33. **Statement-1 :** The moment of inertia of a rigid body reduces to its minimum value as compared to any other parallel axis when the axis of rotation passes through its centre of mass.

Statement-2 : The weight of a rigid body always acts through its centre of mass in uniform gravitational field.

Of these statements:

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

34. **Statement-1 :** A body cannot roll on smooth horizontal surface.

Statement-2 : When a body rolls purely, the point of contact should be at rest with respect to surface.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

35. **Statement-1 :** The angular velocity of all the points on the rigid body as seen from any other point on it is the same.

Statement-2 : The distance between any 2 points on the rigid body remains constant.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

36. **Statement-1 :** KE of rotating rigid body in CM frame is $\frac{1}{2} I_{cm}\omega^2$, where symbols have usual meaning.

Statement-2 : In CM frame rigid body can have only pure rotational motion.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

37. **Statement-1 :** A rigid disc rolls without slipping on a fixed rough horizontal surface with uniform angular velocity. Then the acceleration of lowest point on the disc is zero.

Statement-2 : For a rigid disc rolling without slipping on a fixed rough horizontal surface, the velocity of the lowest point on the disc is always zero.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

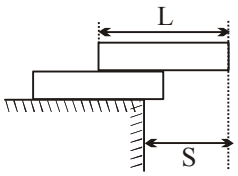
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

38. Two identical bricks of length L are piled one on top of the other on a table as shown in the figure. The maximum distance S the top brick can overhang the table with the system still balanced is

(A) $\frac{1}{2}L$ (B) $\frac{2}{3}L$

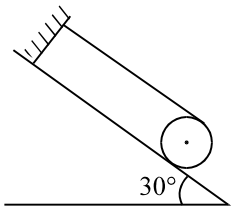
(C) $\frac{3}{4}L$ (D) $\frac{7}{8}L$



39. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure. The minimum coefficient of friction needed for this configuration is

(A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2}$ (D) $\frac{1}{2\sqrt{3}}$



40. A uniform 2 kg cylinder rests on a laboratory cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is closest to the minimum acceleration of the cart needed to cause the cylinder to tip over?

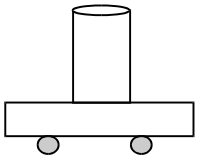
(A) 2 m/s^2

(B) 4 m/s^2

(C) 5 m/s^2

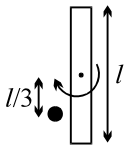
(D) 6 m/s^2

(E) the cylinder would slide at all of these accelerations



41. A uniform rod of length *l* and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance *l*/3 from its axis and stops. Mass of the particle is

(A) 3M (B) $\frac{3M}{4}$ (C) $\frac{3M}{2}$ (D) $\frac{4M}{3}$



42. A yo-yo is released from your hand with the string wrapped around your finger. If you hold your hand still, the acceleration of the yo-yo is

(A) downward, much greater than g (B) downward, much less than g

(C) upward, much less than g (D) upward, much greater than g

(E) downward, at g

43. **Statement-1 :** A cyclist always bends inwards while negotiating a curve
Statement-2 : By bending he lowers his centre of gravity
Of these statements,
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

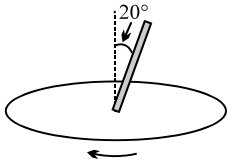
44. **Statement-1 :** $\vec{L} = I\vec{\omega}$ is always true for bodies of all shapes.

Statement-2 : $\vec{\tau} = \dfrac{d\vec{L}}{dt}$ is always true in inertial frames.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Question No. 45 & 46 (2 questions)

A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) If the turntable is rotating clockwise as seen from above.



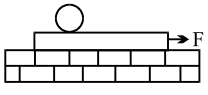
45. What is the direction of the rod's angular momentum vector (calculated about its lower end)?
(A) vertically downwards
(B) down at 20° to the horizontal
(C) up at 20° to the horizontal
(D) vertically upwards

46. Is there a torque acting on it, and if so in what direction?
(A) yes, vertically
(B) yes, horizontally
(C) yes at 20° to the horizontal
(D) no

One or More than one option may be correct

47. A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is v.
(A) $0 \le v \le 2u$
(B) $v = u$, if CP is horizontal
(C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
(D) $v = \sqrt{2}u$, if CP is horizontal

48. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If sphere does not slip over the plank. Which of the following is correct.
(A) Acceleration of the centre of sphere is less than that of the plank.
(B) Work done by friction acting on the sphere is equal to its total kinetic energy.
(C) Total kinetic energy of the system is equal to work done by the force F
(D) None of the above



49. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m. It is rolled down a horizontal plane such that its centre of mass moves with a velocity v. If it purely rolls

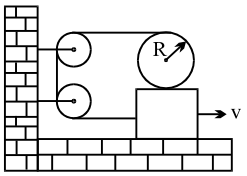
(A) Kinetic energy of the sphere is 5/6 mv^2

(B) Kinetic energy of the sphere is 4/5 mv^2

(C) Angular momentum of the sphere about a fixed point on ground is 8/3 mvR

(D) Angular momentum of the sphere about a fixed point on ground is 14/5 mvR

50. In the figure shown, the plank is being pulled to the right with a constant speed v. If the cylinder does not slip then:



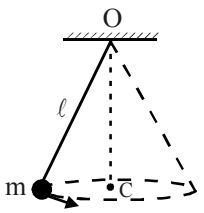
(A) the speed of the centre of mass of the cylinder is 2v.

(B) the speed of the centre of mass of the cylinder is zero.

(C) the angular velocity of the cylinder is v/R.

(D) the angular velocity of the cylinder is zero.

51. A small ball of mass m suspended from the celling at a point O by a thread of length ℓ moves along a horizontal circle with a constant angular velocity ω.



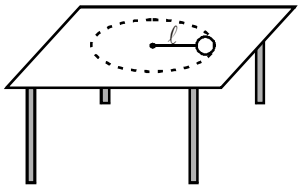
(A) angular momentum is constant about O

(B) angular momentum is constant about C

(C) vertical component of angular momentum about O is constant

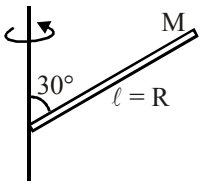
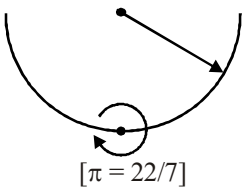
(D) Magnitude of angular momentum about O is constant

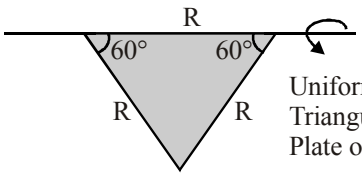
52. A disc is moving on a rough horizontal table in a circular path being tied to a nail by a string. While the disc is in motion, match the quantities in column-I with their description in column-II.



	Column-I		**Column-II**
(A)	Acceleration of disc	(P)	Changes direction but constant in magnitude
(B)	Friction force on disc	(Q)	Changes magnitude but constant in direction
(C)	Angular momentum of the particle	(R)	Changes direction as well as magnitude
(D)	Normal force by table	(S)	Direction as well as the magnitude is constant

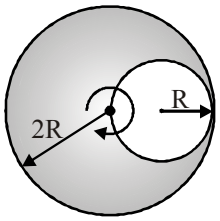
53.

	Column-I		**Column-II**
(A)	 Uniform Rod	(P)	$\frac{8MR^2}{11}$
(B)	 Uniform Semicircular Ring. Axis is perpendicular to plane of ring.	(Q)	$\frac{MR^2}{12}$

(C) 

Uniform Triangular Plate of mass M

(R) $\frac{13MR^2}{8}$

(D) 

Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate

(S) $\frac{MR^2}{8}$

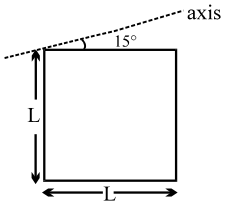
54. An object is allowed to roll down the incline starting from rest. All are uniform and have same mass and radius

	Column-I		**Column-II**
(A)	The object which has largest rotational inertia about its axes of symmetry	(P)	Solid sphere
(B)	The object which will experience the largest net torque about CM	(Q)	Spherical shell
(C)	The object which will have the largest speed at the bottom of the incline	(R)	Solid disk
(D)	The object which will reach the bottom of incline in the shortest time.	(S)	Thin hollow cylinder

EXERCISE-II

1. Two rods of equal mass m and length *l* lie along the x axis and y axis with their centres origin. What is the moment of inertia of both about the line x = y :

2. A square plate of mass M and edge L is shown in figure. Find the moment of inertia of the plate about the axis in the plane of plate passing through one of its vertex making an angle 15° from horizontal.



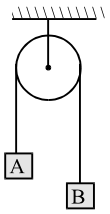
3. A cube is in equilibrium on an inclined plane forming an angle of 30° with the horizontal. What is distance of line of action of the normal reaction of the plane on the cube from the centre.

4. In an experiment with a beam balance on unknown mass m is balanced by two known mass m is balanced by two known masses of 16 kg and 4 kg as shown in figure. (massless beam)

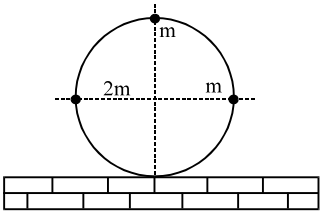


Find the value of the unknown mass m.

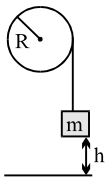
5. In the figure A & B are two blocks of mass 4 kg & 2 kg respectively attached to the two ends of a light string passing over a disc C of mass 40 kg and radius 0.1 m. The disc is free to rotate about a fixed horizontal axes, coinciding with its own axis. The system is released from rest and the string does not slip over the disc. Find :
(i) the linear acceleration of mass B .
(ii) the number of revolutions made by the disc at the end of 10 sec. from the start.
(iii) the tension in the string segment supporting the block A.



6. A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0. Find the kinetic energy of the system. (Slipping is absent)

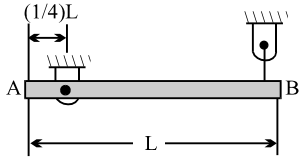


7. A mass m is attached to a pulley through a cord as shown in the fig. The pulley is a solid disk with radius R. The cord does not slip on the disk. The mass is released from rest at a height h from the ground and at the instant the mass reaches the ground, the disk is rotating with angular velocity ω. Find the mass of the disk.



8. A solid homogeneous cylinder of height h and base radius r is kept vertically on a conveyer belt moving horizontally with an increasing velocity $v = a + bt^2$. If the cylinder is not allowed to slip find the time when the cylinder is about to topple.

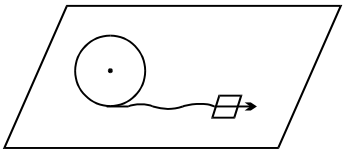
9. A uniform beam of length L and mass m is supported as shown. If the cable suddenly breaks, determine ;
(a) the acceleration of end B.
(b) the reaction at the pin support.



10. A solid uniform disk of mass m rolls without slipping down a fixed inclined plane with an acceleration a. Find the frictional force on the disk due to surface of the plane.

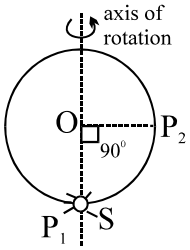
11. A ring of radius R rolls without sliding with a constant velocity. Find the radius of curvature of the path followed by any particle of the ring at the highest point of its path.

12. A block of mass m is attached to a pulley disc of equal mass m, radius r by means of a slack string as shown. The pulley is hinged about its centre on a horizontal table and the block is projected with an initial velocity of 5 m/s. Find the velocity when the string becomes taut.



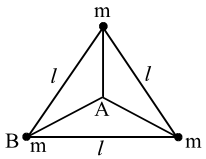
13. The angular momentum of a flywheel having a moment of inertia of 0.4 kg m^2 decreases from 30 to 20 kg m^2/s in a period of 2 second . Find the average torque acting on the flywheel during this period.

14. A uniform ring is rotating about vertical axis with angular velocity ω initially. A point insect (S) having the same mass as that of the ring starts walking from the lowest point P_1 and finally reaches the point P_2(as shown in figure). What is the final angular velocity of the ring?

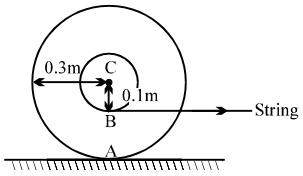


15. A particle of mass 10 kg is moving with a uniform speed of 6 m/sec. in x–y plane along the line 3y = 4x + 10, what is the magnitude of its angular momentum about the origin in kg–m^2/s?

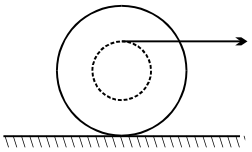
16. Three equal masses m are rigidly connected to each other by massless rods of length *l* forming an equilateral triangle, as shown above. The assembly is to be given an angular velocity ω about an axis perpendicular to the triangle. For fixed ω, what is the ratio of the kinetic energy of the assembly for an axis through B compared with that for an axis through A.



17. A wheel is made to roll without slipping, towards right, by pulling a string wrapped around a coaxial spool as shown in figure. With what velocity the string should be pulled so that the centre of the wheel moves with a velocity of 3 m/s?

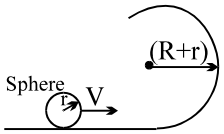


18. A spool of inner radius R and outer radius 3R has a moment of inertia = MR^2 about an axis passing through its geometric centre, where M is the mass of the spool. A thread woudn on the inner surface of the spool is pulled horizontally with a constant force = Mg.

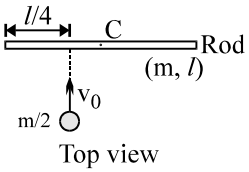


Find the acceleration of the point on the thread which is being pulled assuming that the spool rolls purely on the floor.

19. A sphere of mass m and radius r is pushed onto the fixed horizontal surface such that it rolls without slipping from the beginning. Determine the minimum speed v of its mass centre at the bottom so that it rolls completely around the loop of radius (R + r) without leaving the track in between.

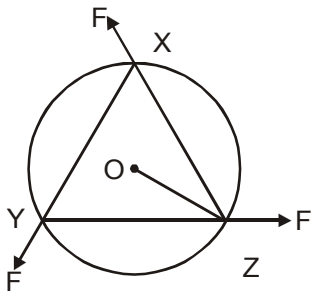


20. A uniform rod of mass m and length *l* is resting on a smooth horizontal surface. A particle of mass m/2 travelling with a speed v_0 hits the rod normally and elastically. Find final velocity of particle and the angular velocity of the rod.

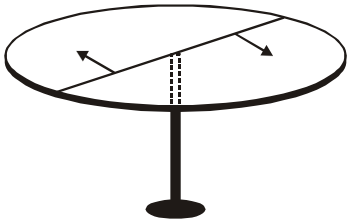


EXERCISE-III

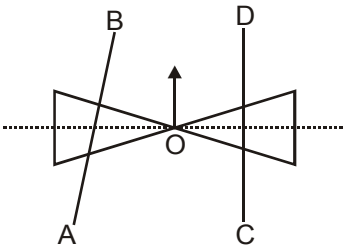
1. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude F = 0.5 N are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is **[JEE (Advanced)-2014]**



2. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is **[JEE (Advanced)-2014]**

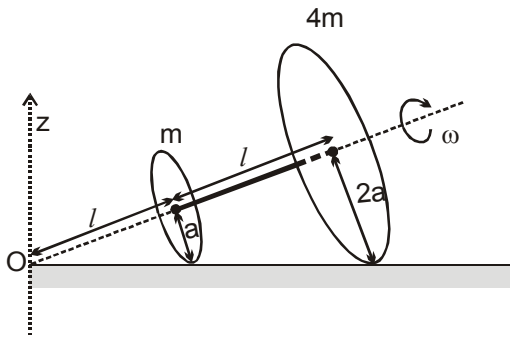


3. A roller is made by joining together two cones at their vertices O. It is kept on two rails AB and CD which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to **[JEE (Mains)-2016]**



(A) Turn left
(B) Turn right
(C) Go straight
(D) Turn left and right alternately

4. Two thin circular discs of mass m and 4m, having radii of a and 2a, respectively, are rigidly fixed by a massless, rigid rod of length $l = \sqrt{24}$ a through their centers. This assembly is laid on a firm and flat surace, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω. The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure), Which of the following statement(s) is(are) true? **[JEE Advance 2016]**



(A) The magnitude of angular momentum of the assembly about its center of mass is 17 $ma^2\omega/2$.

(B) The magnitude of the z-component of $\vec{L}$ is $55ma^2\omega$.

(C) The magnitude of angular momentum of center of mass of the assembly about the point O is $81ma^2\omega$.

(D) The center of mass of the assembly rotates about the z-axis with an angular speed of $\omega/5$.

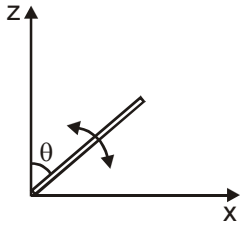
5. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I. What is the ratio ℓ / R such that the moment of inertia is minimum **[JEE Main 2017]**

(A) $\frac{\sqrt{3}}{2}$ (B) 1 (C) $\frac{3}{\sqrt{2}}$ (D) $\sqrt{\frac{3}{2}}$

6. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is **[JEE Main 2017]**

(A) $\frac{2g}{3\ell}\sin\theta$ (B) $\frac{3g}{2\ell}\cos\theta$

(C) $\frac{2g}{3\ell}\cos\theta$ (D) $\frac{3g}{2\ell}\sin\theta$



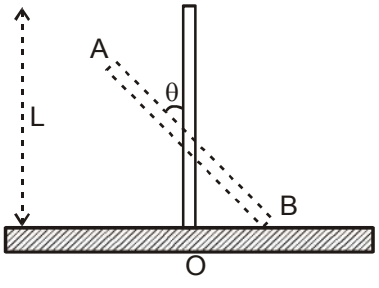
7. A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle mde by the bar with the vertical is θ. Which of the following statements about its motion is/are correct? **[JEE Advance 2017]**

(A) When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to $(1 - \cos\theta)$

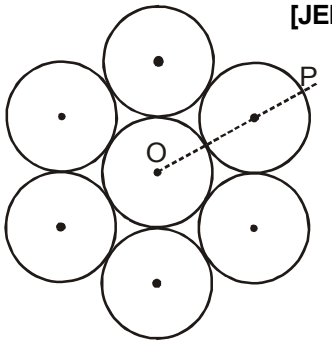
(B) The midpoint of the bar will fall vertically downward

(C) Instantaneous torque about the point in contact with the floor is proportional to $\sin\theta$

(D) The trajectory of the point A is a parabola



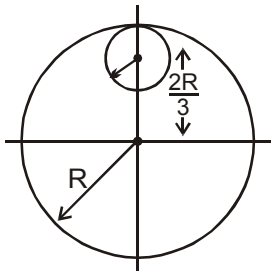
8. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is : **[JEE-Mains-2018]**



(A) $\frac{181}{2}MR^2$ (B) $\frac{19}{2}MR^2$

(C) $\frac{55}{2}MR^2$ (D) $\frac{73}{2}MR^2$

9. From a uniform circular disc of radius R and mass 9 M, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is : **[JEE-Mains-2018]**



(A) $\frac{37}{9}MR^2$ (B) 4 MR^2

(C) $\frac{40}{9}MR^2$ (D) 10 MR^2

10. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O. If v is the speed of the particle and L is the magnitude of its angular momentum about O, which of the following statements is (are) true? **[JEE-Advanced-2018]**

(A) $v = \sqrt{\frac{k}{2m}}R$ (B) $v = \sqrt{\frac{k}{m}}R$ (C) $L = \sqrt{mk}\ R^2$ (D) $L = \sqrt{\frac{mk}{2}}\ R^2$

11. A ring and a disc are initially at rest, side by side, at the top of an inclined plane which makes an angle 60º with the horizontal. They start to roll without slipping at the same instant of time along the shortest path. If the time difference between their reaching the ground is $\frac{(2-\sqrt{3})}{\sqrt{10}}$ s, then the height of the top of the inclined plane, in metres, is ..... Take g = 10 m s^{-2}. **[JEE-Advanced-2018]**

12. A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its center. The moment of inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = a MR^2$. The value of the coefficient a is **[JEE-Mains-2019]**

(1) $\frac{8}{5}$ (2) $\frac{3}{5}$ (3) $\frac{1}{2}$ (4) $\frac{3}{2}$

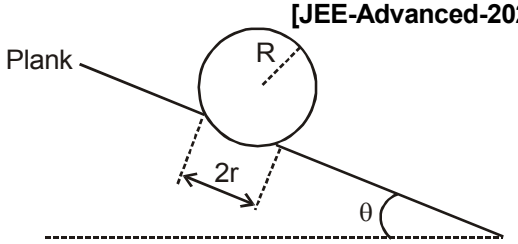
13. The radius of gyration of a uniform rod of length l, about an axis passing through a point $\frac{l}{4}$ away from the centre of the rod, and perpendicular to it, is : **[JEE-Mains-2020]**

(1) $\sqrt{\frac{3}{8}}\,l$ (2) $\sqrt{\frac{7}{48}}\,l$ (3) $\frac{1}{4}l$ (4) $\frac{1}{8}l$

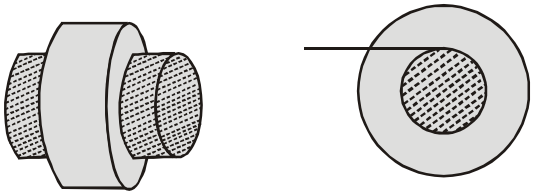
14. A football of radius R is kept on a hole of radius r (r < R) made on a plank kept horizontally. One end of the plank is now lifted so that it gets tilted making an angle θ from the horizontal as shown in the figure below. The maximum value of θ so that the football does not start rolling down the plank satisfies (figure is schematic and not drawn to scale) **[JEE-Advanced-2020]**

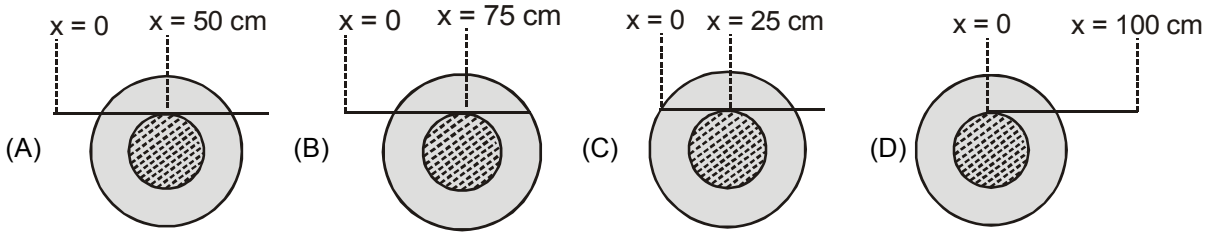
(A) $\sin\theta = \frac{r}{R}$ (B) $\tan\theta = \frac{r}{R}$

(C) $\sin\theta = \frac{r}{2R}$ (D) $\cos\theta = \frac{r}{2R}$



15. A small roller of diameter 20 cm has an axle of diameter 10 cm (see figure below on the left). It is on a horizontal floor and a meter scale is positioned horizontally on its axle with one edge of the scale on top of the axle (see figure on the right). The scale is now pushed slowly on the axle so that it moves without slipping on the axle, and the roller starts rolling without slipping. After the roller has moved 50 cm, the position of the scale will look like (figures are schematic and not drawn to scale) **[JEE-Advanced-2020]**





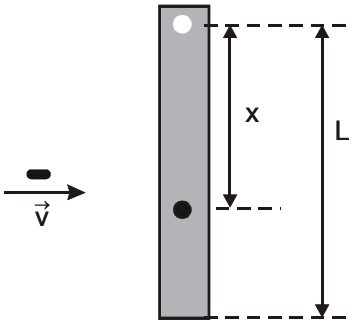
16. A rod of mass m and length L, pivoted at one of its ends, is hanging vertically. A bullet of the same mass moving at speed v strikes the rod horizontally at a distance x from its pivoted end and gets embedded in it. The combined system now rotates with angular speed ω about the pivot. The maximum angular speed ω_M is achieved for $x = x_M$. Then **[JEE-Advanced-2020]**

(A) $\omega = \frac{3vx}{L^2 + 3x^2}$

(B) $\omega = \frac{12vx}{L^2 + 12x^2}$

(C) $x_M = \frac{L}{\sqrt{3}}$

(D) $\omega_M = \frac{v}{2L}\sqrt{3}$



17. Two discs have moments of intertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by : **[JEE-Mains-2021]**

(A) $\frac{(\omega_1 - \omega_2)^2}{2I_1 + I_2}$ (B) $\frac{(I_1 - I_2)^2 \omega_1 \omega_2}{2(I_1 + I_2)}$ (C) $\frac{I_1 I_2}{2(I_1 + I_2)}(\omega_1 - \omega_2)^2$ (D) $\frac{I_1 I_2}{(I_1 + I_2)}(\omega_1 - \omega_2)^2$

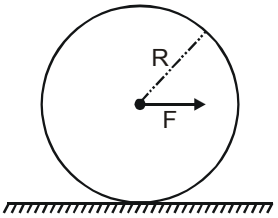
18. A horizontal force F is applied at the center of mass of a cylindrical object of mass m and radius R, perpendicular to its axis as shown in the figure. The coefficient of friction between the object and the ground is μ. The center of mass of the object has an acceleration a. The acceleration due to gravity is g. Given that the object rolls without slipping, which of the following statement(s) is(are) correct? **[JEE-Advanced-2021]**

(A) For the same F, the value of a does not depend on whether the cylinder is solid or hollow

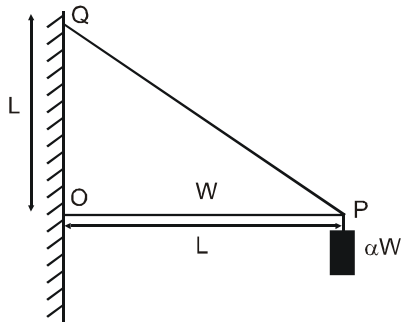
(B) For a solid cylinder, the maximum possible value of a is $2\mu g$

(C) The magnitude of the frictional force on the object due to the ground is always μmg

(D) For a thin-walled hollow cylinder, $a = \frac{F}{2m}$



19. One end of a horizontal uniform beam of weight W and length L is hinged on a vertical wall at point O and its other end is supported by a light inextensible rope. The other end of the rope is fixed at point Q, at a height L above the hinge at point O. A block of weight αW is attached at the point P of the beam, as shown in the figure (not to scale). The rope can sustain a maximum tension of $(2\sqrt{2})$ W. Which of the following statement(s) is(are) correct? **[JEE-Advanced-2021]**



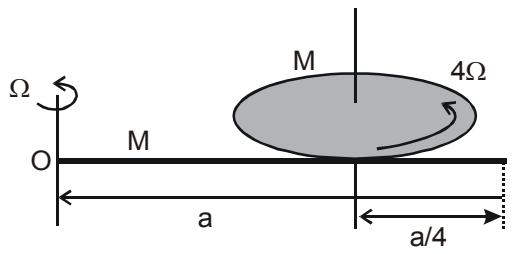
(A) The vertical component of reaction force at O does **not** depend on α

(B) The horizontal component of reaction force at O is equal to W for $\alpha = 0.5$

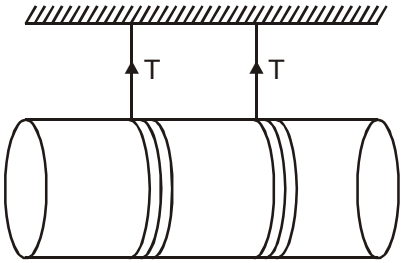
(C) The tension in the rope is 2W for $\alpha = 0.5$

(D) The rope breaks if $\alpha > 1.5$

20. A thin rod of mass M and length a is free to rotate in horizontal plane about a fixed vertical axis passing through point O. A thin circular disc of mass M and of radius a/4 is pivoted on this rod with its center at a distance a/4 from the free end so that it can rotate freely about its vertical axis, as shown in the figure. Assume that both the rod and the disc have uniform density and they remain horizontal during the motion. An outside stationary observer finds the rod rotating with an angular velocity Ω and the disc rotating about its vertical axis with angular velocity 4Ω. The total angular momentum of the system about the point O is $\left(\frac{Ma^2\Omega}{48}\right)n$. The value of n is _____. **[JEE-Advanced-2021]**



21. A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should by unbinding the strings to achieve a speed of 4 ms^{-1}, is…….cm. (take g = 10 ms^{-2}) **[JEE-Mains-2022]**



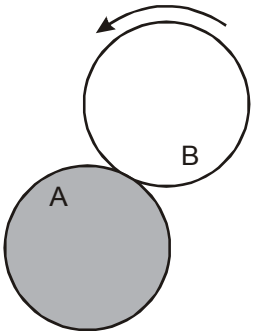
22. A flat surface of a thin uniform disk A of radius R is glued to a horizontal table. Another thin uniform disk B of mass M and with the same radius R rolls without slipping on the circumference of A, as shown in the figure. A flat surface of B also lies on the plane of the table. The center of mass of B has fixed angular speed ω about the vertical axis passing through the center of A. The angular momentum of B is $nM\omega R^2$ with respect to the center of A. Which of the following is the value of n? **[JEE-Advanced-2022]**

(A) 2

(B) 5

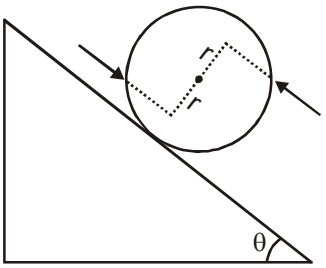
(C) $\frac{7}{2}$

(D) $\frac{9}{2}$



23. At time t = 0, a disk of radius 1 m starts to roll without slipping on a horizontal plane with an angular acceleration of $\alpha = \frac{2}{3}$ rad s^{-2}. A small stone is stuck to the disk. At t = 0, it is at the contact point of the disk and the plane. Later, at time $t = \sqrt{\pi}\,s$, the stone detaches itself and flies off tangentially from the disk. The maximum height (in m) reached by the stone measured from the plane is $\frac{1}{2} + \frac{x}{10}$. The value of x is _____. [Take g = 10 ms^{-2}] **[JEE-Advanced-2022]**

24. A solid sphere of mass 1 kg and radius 1 m rolls without slipping on a fixed inclined plane with an angle of inclination θ = 30° from the horizontal. Two forces of magnitude 1 N each, parallel to the incline, act on the sphere, both at distance r = 0.5 m from the center of the sphere, as shown in the figure. The acceleration of the sphere down the plane is ____ ms^{-2}. (Take g = 10 ms^{-2}). **[JEE-Advanced-2022]**



25. Two identical solid spheres each of mass 2 kg and radii 10 cm are fixed at the ends of a light rod. The separation between the centres of the spheres is 40 cm. The moment of inertia of the system about an axis perpendicular to the rod passing through its middle point _____ is × 10^{-3} kg-m^2. **[JEE-Mains-2023]**

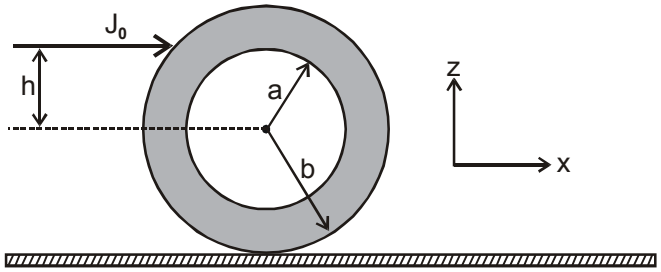
26. A bar of mass M = 1.00 kg and length L = 0.20 m is lying on a horizontal frictionless surface. One end of the bar is pivoted at a point about which it is free to rotate. A small mass m = 0.10 kg is moving on the same horizontal surface with 5.00 m s^{-1} speed on a path perpendicular to the bar. It hits the bar at a distance L/2 from the pivoted end and returns back on the same path with speed v. After this elastic collision, the bar rotates with an angular velocity ω. Which of the following statement is correct? **[JEE-Advanced-2023]**

(A) ω = 6.98 rad s^{-1} and v = 4.30 m s^{-1} (B) ω = 3.75 rad s^{-1} and v = 4.30 m s^{-1}

(C) ω = 3.75 rad s^{-1} and v = 10.0 m s^{-1} (D) ω = 6.80 rad s^{-1} and v = 4.10 m s^{-1}

27. An annular disk of mass M, inner radius a and outer radius b is placed on a horizontal surface with coefficient of friction μ, as shown in the figure. At some time, an impulse $J_0\hat{x}$ is applied at a height h above the center of the disk. If $h = h_m$ then the disk rolls without slipping along the x-axis.

Which of the following statement(s) is(are) correct? **[JEE-Advanced-2023]**



(A) For $\mu \neq 0$ and $a \to 0$, $h_m = b/2$

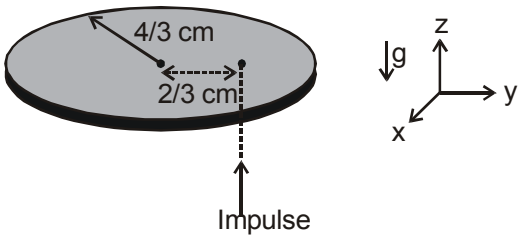
(B) For $\mu \neq 0$ and $a \to b$, $h_m = b$

(C) For $h = h_m$, the initial angular velocity does **not** depend on the inner radius a.

(D) For $\mu = 0$ and h = 0, the wheel always slides without rolling.

28. A thin circular coin of mass 5 gm and radius 4/3 cm is initially in a horizontal xy-plane. The coin is tossed vertically up (+z direction) by applying an impulse of $\sqrt{\frac{\pi}{2}} \times 10^{-2}$ N-s at a distance 2/3 cm from its center. The coin spins about its diameter and moves along the +z direction. By the time the coin reaches back to its initial position, it completes n rotations. The value of n is ____.

[Given : The acceleration due to gravity g = 10 m s^{-2}] **[JEE-Advanced-2023]**



29. A thin circular disc of mass m and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with angular velocity ω. If another disc of same dimensions but of mass M/2 is placed gently on the first disc co-axially, then the new angular velocity of the system is :

(A) $\frac{2}{3}\omega$ (B) $\frac{4}{5}\omega$ (C) $\frac{5}{4}\omega$ (D) $\frac{3}{2}\omega$ **[JEE-Mains-2024]**

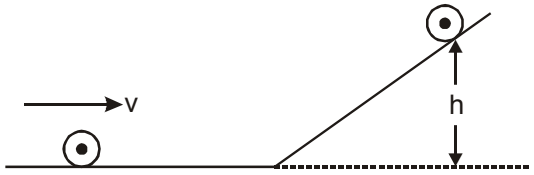
30. A heavy iron bar of weight 12 kg is having its one end on the ground and the other on the shoulder of a man. The rod makes an angle 60° with the horizontal, the weight experienced by the man is **[JEE-Mains-2024]**

(A) 12 kg (B) $6\sqrt{3}$ kg (C) 3 kg (D) 6 kg

31. A particle of mass m is projected with a velocity 'u' making an angle of 30º with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height h is : **[JEE-Mains-2024]**

(A) $\frac{mu^3}{\sqrt{2}g}$ (B) $\frac{\sqrt{3}}{16}\frac{mu^3}{g}$ (C) zero (D) $\frac{\sqrt{3}}{2}\frac{mu^2}{g}$

32. A disc of radius R and mass M is rolling horizontally without slipping with speed v. It then moves up an inclined smooth surface as shown in figure. The maximum height that the disc can go up the incline is



(A) $\frac{2}{3}\frac{v^2}{g}$ (B) $\frac{v^2}{g}$ (C) $\frac{3}{4}\frac{v^2}{g}$ (D) $\frac{1}{2}\frac{v^2}{g}$ **[JEE-Mains-2024]**

33. A body of mass 'm' is projected with a speed 'u' making an angle of 45º with the ground. The angular momentum of the body about the point of projection, at the highest point is expressed as $\frac{\sqrt{2}mu^3}{Xg}$. The value of 'X' is _____. **[JEE-Mains-2024]**

34. Consider a Disc of mass 5 kg, radius 2m, rotating with angular velocity of 10 rad/s about an axis perpendicular to the plane of rotation. An identical disc is kept gently over the rotating disc along the same axis. The energy dissipated so that both the discs continue to rotate together without slipping is_____J. **[JEE-Mains-2024]**

35. Two identical spheres each of mass 2kg and radius 50 cm are fixed at the ends of a light rod so that the separation between the centers is 150 cm. Then, moment of inertia of the system about an axis perpendicular to the rod and passing through its middle point is $\frac{x}{20}$ kg m^2, where the value of x is____. **[JEE-Mains-2024]**

36. A ring and a solid sphere roll down the same inclined plane without slipping. They start from rest. The radii of both bodies are identical and the ratio of their kinetic energies is $\frac{7}{x}$, where x is_____. **[JEE-Mains-2024]**

37. A body of mass 5 kg moving with a uniform speed $3\sqrt{2}$ ms^{-1} in X-Y plane along the line y = x + 4. The angular momentum of the particle about the origin will be _____ kg m^2s^{-1}. **[JEE-Mains-2024]**

38. Two discs of moment of inertian $I_1 = 4$ kg m^2 and $I_2 = 2$ kg m^2, about their central axes and normal to their planes, rotating with angular speeds 10 rad/s and 4 rad/s respectively are brought into contact face to face with their axes of rotation coincident. The loss in kinetic energy of the system in the process is _____J. **[JEE-Mains-2024]**

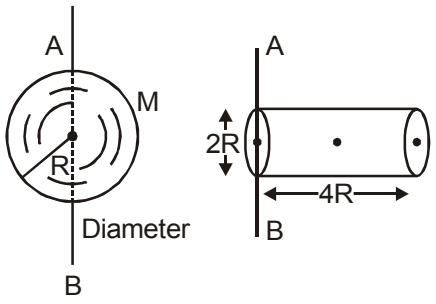
39. A solid circular disc of mass 50 kg rolls along a horizontal floor so that its center of mass has a speed of 0.4 m/s. The absolute value of work done on the disc to stop it is _____ J. **[JEE-Mains-2024]**

40. Four particles each of mass 1 kg are placed at four corners of a square of side 2m. Moment of inertia of system about an axis perpendicular to its plane and passing through one of its vertex is _____ kgm^2. **[JEE-Mains-2024]**

41. A cylinder is rolling down on an inclined plane of inclination 60º. It's acceleration during rolling down will be $\frac{x}{\sqrt{3}}\ m/s^2$, where x = _____(use g = 10 m/s^2). **[JEE-Mains-2024]**

42. A uniform rod AB of mass 2 kg and length 30 cm at rest on a smooth horizontal surface. An impulse of force 0.2 Ns is applied to end B. The time taken by the rod to turn through at right angles will be $\frac{\pi}{x}$ s, where x = _____. **[JEE-Mains-2024]**

43. Ratio of radius of gyration of a hollow sphere to that of a solid cylinder of equal mass, for moment of inertia about their diameter axis AB as shown in figure is $\sqrt{\frac{8}{x}}$. The value of x is **[JEE-Mains-2024]**



(A) 51 (B) 17 (C) 67 (D) 34

44. A solid sphere and a hollow cylinder roll up without slipping on same inclined plane with same initial speed υ. The sphere and the cylinder reaches upto maximum heights h_1 and h_2, respectively, above the initial level. The ratio $h_1 : h_2$ is $\frac{n}{10}$. The value of n is _____. **[JEE-Mains-2024]**

45. A hollow sphere is rolling on a plane surface about its axis of symmetry. The ratio of rotational kinetic energy to its total kinetic energy is $\frac{x}{5}$. The value of x is_____. **[JEE-Mains-2024]**

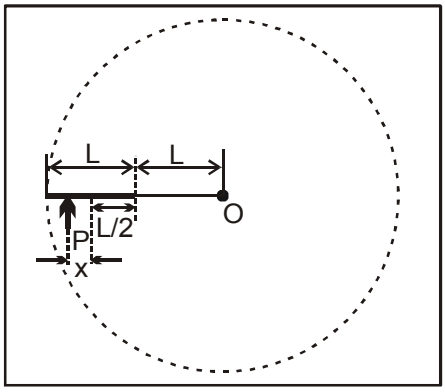
46. If the radius of earth is reduced to three-fourth of its present value without change in its mass then value of duration of the day of earth will be_____ hours 30 minutes. **[JEE-Mains-2024]**

47. Three balls of masses 2kg, 4kg and 6kg respectively are arranged at centre of the edges of an equilateral triangle of side 2 m. The moment of intertia of the system about an axis through the centroid and perpendicular to the plane of triangle, will be _____ kg m^2. **[JEE-Mains-2024]**

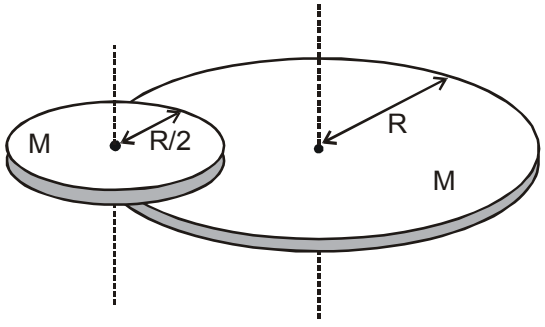
48. A string is wrapped around the rim of a wheel of moment of inertia 0.40 kgm^2 and radius 10 cm. The wheel is free to rotate about its axis initially the wheel is at rest. The string is now pulled by a force of 40 N. The angular velocity of the wheel after 10 s is x rad/s, where x is _____. **[JEE-Mains-2024]**

49. A circular disc reaches from top to bottom of an inclined plane of length l. When it slipsdown the plane, if takes t s. When it rolls down the plane then it takes $\left(\frac{\alpha}{2}\right)^{1/2}$ t s, where α is _____. **[JEE-Mains-2024]**

50. A thin uniform rod of length L and certain mass is kept on a frictionless horizontal table with a massless string of length L fixed to one end (top view is shown in the figure). The other end of the string is pivoted to a point O. If a horizontal impulse P is imparted to the rod at a distance x = L/n from the mid-point of the rod (see figure), then the rod and string revolve together around the point O, with the rod remaining aligned with the string. In such a case, the value of n is _____. **[JEE-Advanced-2024]**



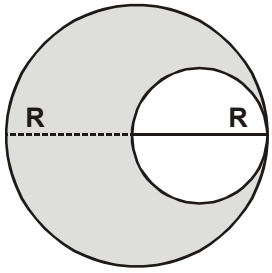
51. A disc of mass M and radius R is free to rotate about its vertical axis as shown in the figure. A battery operated motor of negligible mass is fixed to this disc at a point on its circumference. Another disc of the same mass M and radius R/2 is fixed to the motor's thin shaft. Initially, both the discs are at rest. The motor is switched on so that the smaller disc rotates at a uniform angular speed ω. If the angular speed at which the large disc rotates is ω/n, then the value of n is _____. **[JEE-Advanced-2024]**



52. The moment of inertia of a circular ring of mass M and diameter r about a tangential axis lying in the plane of the ring is : **[JEE-Mains-2025]**

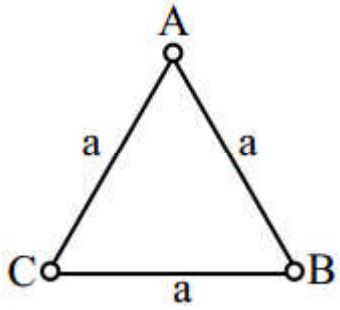
(A) $\frac{1}{2}Mr^2$ (B) $\frac{3}{8}Mr^2$ (C) $\frac{3}{2}Mr^2$ (D) $2\,Mr^2$

53. A uniform circular disc of radius 'R' and mass 'M' is rotating about an axis perpendicular to its plane and passing through its centre. A small circular part of radius R/2 is removed from the original disc as shown in the figure. Find the moment of inertia of the remaining part of the original disc about the axis as given above.



(A) $\frac{7}{32}MR^2$ (B) $\frac{9}{32}MR^2$ (C) $\frac{17}{32}MR^2$ (D) $\frac{13}{32}MR^2$ **[JEE-Mains-2025]**

54. Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At t = 0, they are given an initial velocity $\vec{V}_A = V_0\overrightarrow{AC}$, $\vec{V}_B = V_0\overrightarrow{BA}$ and $\vec{V}_C = V_0\overrightarrow{CB}$. Here, $\overrightarrow{AC}$, $\overrightarrow{CB}$ and $\overrightarrow{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is : **[JEE-Mains-2025]**



(A) $\frac{1}{2}amV_0$ (B) $3amV_0$ (C) $\frac{\sqrt{3}}{2}amV_0$ (D) $\frac{3}{2}amV_0$

55. A solid sphere and a hollow sphere of the same mass and of same radius are rolled on an inclined plane. Let the time taken to reach the bottom by the solid sphere and the hollow sphere be t_1 and t_2, respectively, then

(A) $t_1 < t_2$ (B) $t_1 = t_2$ (C) $t_1 = 2t_2$ (D) $t_1 > t_2$ **[JEE-Mains-2025]**

56. A uniform rod of mass 250 g having length 100 cm is balanced on a sharp edge at 40 cm mark. A mass of 400 g is suspended at 10 cm mark. To maintain the balance of the rod, the mass to be suspended at 90 cm mark, is **[JEE-Mains-2025]**

(A) 300 g (B) 190 g (C) 200 g (D) 290 g

57. A circular disk of radius R meter and mass M kg is rotating around the axis perpendicular to the disk. An external torque is applied to the disk such that $\theta(t) = 5t^2 - 8t$, where $\theta(t)$ is the angular position of the rotating disc as a function of time t. **[JEE-Mains-2025]**

How much power is delivered by the applied torque, when t = 2s?

(A) 60 MR^2 (B) 72 MR^2 (C) 108 MR^2 (D) 8 MR^2

58. A solid sphere is rolling without slipping on a horizontal plane. The ratio of the linear kinetic energy of the centre of mass of the sphere and rotational kinetic energy is **[JEE-Mains-2025]**

(A) $\frac{2}{5}$ (B) $\frac{5}{2}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

59. A uniform solid cylinder of mass 'm' and radius 'r' rolls along an inclined rough plane of inclination 45°. If it starts to roll from rest from the top of the plane then the linear acceleration of the cylinder axis will be

(A) $\frac{1}{\sqrt{2}}g$ (B) $\frac{1}{3\sqrt{2}}g$ (C) $\frac{\sqrt{2}g}{3}$ (D) $\sqrt{2}g$ **[JEE-Mains-2025]**

60. A solid sphere of mass 'm' and radius 'r' is allowed to roll without slipping from the highest point of an inclined plane of length 'L' and makes an angle 30º with the horizontal. The speed of the particle at the bottom of the plane is v_1. If the angle of inclination is increased to 45º while keeping L constant. Then the new speed of the sphere at the bottom of the plane is v_2. The ratio of $v_1^2 : v_2^2$ is **[JEE-Mains-2025]**

(A) $1 : \sqrt{2}$ (B) 1 : 3 (C) 1 : 2 (D) $1 : \sqrt{3}$

61. An object of mass 'm' is projected from origin in a vertical xy plane at an angle 45° with the x-axis with an initial velocity v_0. The magnitude and direction of the angular momentum of the object with respect to origin, when it reaches at the maximum height, will be [g is acceleration due to gravity] **[JEE-Mains-2025]**

(A) $\frac{mv_0^3}{2\sqrt{2}g}$ along negative z-axis (B) $\frac{mv_0^3}{2\sqrt{2}g}$ along positive z-axis

(C) $\frac{mv_0^3}{4\sqrt{2}g}$ along positive z-axis (D) $\frac{mv_0^3}{4\sqrt{2}g}$ along negative z-axis

62. The torque due to the force $(2\hat{i} + \hat{j} + 2\hat{k})$ about the origin, acting on a particle whose position vector is $(\hat{i} + \hat{j} + \hat{k})$, would be : **[JEE-Mains-2025]**

(A) $\hat{i} - \hat{j} + \hat{k}$ (B) $\hat{j} + \hat{k}$ (C) $\hat{i} - \hat{k}$ (D) $\hat{j} - \hat{k}$

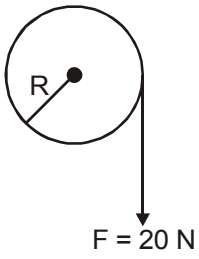
63. The position vectors of two 1 kg particles, (A) and (B), are given by $\vec{r}_A = \left(\alpha_1 t^2\hat{i} + \alpha_2 t\hat{j} + \alpha_3 t\hat{k}\right)m$ and $\vec{r}_B = \left(\beta_1 t\hat{i} + \beta_2 t^2\hat{j} + \beta_3 t\hat{k}\right)m$, respectively ; ($\alpha_1$ = 1 m/s², α_2 = 3n m/s, α_3 = 2 m/s, β_1 = 2 m/s, β_2 = –1 m/s², β_3 = 4p m/s), where t is time, n and p are constants, At t = 1s, $|\vec{V}_A| = |\vec{V}_B|$ and velocities $\vec{V}_A$ and $\vec{V}_B$ of the particles are orthogonal to each other. At t = 1 s, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{L}$ kgm^2s^{-1}. The value of L is _____. **[JEE-Mains-2025]**

64. Two iron solid discs of negligible thickness have radii R_1 and R_2 and moment of intertia I_1 and I_2, respectively. For $R_2 = 2R_1$, the ratio of I_1 and I_2 would be 1/x, where x = _____ **[JEE-Mains-2025]**

65. The moment of inertia of a solid disc rotating along its diameter is 2.5 times higher than the moment of inertia of a ring rotating in similar way. The moment of inertia of a solid sphere which has same radius as the disc and rotating in similar way, is n times higher than the moment of inertia of the given ring. Here, n =_____. (Consider all the bodies have equal masses). **[JEE-Mains-2025]**

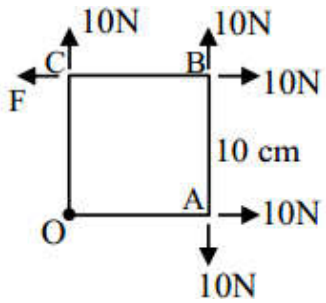
66. The coordinates of a particle with respect to origin in a given reference frame is (1, 1, 1) meters. If a force of $\vec{F} = \hat{i} - \hat{j} + \hat{k}$ acts on the particle, then the magnitude of torque (with respect to origin) in z-direction is_____. **[JEE-Mains-2025]**

67. A cord of negligible mass is wound around the rim of a wheel supported by spokes with negligible mass. The mass of wheel is 10 kg and radius is 10 cm and it can freely rotate without any friction. Initially the wheel is at rest. If a steady pull of 20 N is applied on the cord, the angular velocity of the wheel, after the cord is unwound by 1 m, would be **[JEE-Mains-2025]**



(A) 20 rad/s (B) 30 rad/s (C) 10 rad/s (D) 0 rad/s

68. A square Lamina OABC of length 10 cm is pivoted at 'O'. Forces act at Lamina as shown in figure. If Lamina remains stationary, then the magnitude of F is : **[JEE-Mains-2025]**

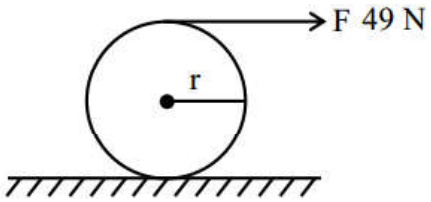


(A) 20 N (B) 0 (zero) (C) 10 N (D) $10\sqrt{2}$ N

69. Moment of inertia of a rod of mass 'M' and length 'L' about an axis passing through its center and normal to its length is 'α'. Now the rod is cut into two equal parts and these parts are joined symmetrically to form a cross shape. Moment of inertia of cross about an axis passing through its center and normal to plane containing cross is : **[JEE-Mains-2025]**

(A) α (B) $\alpha/4$ (C) $\alpha/8$ (D) $\alpha/2$

70. A force of 49 N acts tangentially at the highest point of a sphere (solid) of mass 20 kg, kept on a rough horizontal plane. If the sphere rolls without slipping, then the acceleration of the center of the sphere is



(A) 3.5 m/s^2 (B) 0.35 m/s^2 (C) 2.5 m/s^2 (D) 0.25 m/s^2 **[JEE-Mains-2025]**

71. A wheel is rolling on a plane surface. The speed of a particle on the highest point of the rim is 8 m/s. The speed of the particle on the rim of the wheel at the same level as the centre of wheel, will be **[JEE-Mains-2025]**

(A) $4\sqrt{2}$ m / s (B) 8 m/s (C) 4 m/s (D) $8\sqrt{2}$ m / s

72. Which of the following are correct expression for torque acting on a body? **[JEE-Mains-2025]**

(A) $\vec{\tau} = \vec{r} \times \vec{L}$ (B) $\vec{\tau} = \frac{d}{dt}(\vec{r} \times \vec{p})$ (C) $\vec{\tau} = \vec{r} \times \frac{d\vec{p}}{dt}$ (D) $\vec{\tau} = I\vec{\alpha}$ (E) $\vec{\tau} = \vec{r} \times \vec{F}$

($\vec{r}$ = position vector; $\vec{p}$ = linear momentum; $\vec{L}$ = angular momentum; $\vec{\alpha}$ = angular acceleration; I = moment of inertia; $\vec{F}$ = force; t = time)

Choose the correct answer from the options given below :

(A) B, D and E Only (B) C and D Only

(C) B, C, D and E Only (D) A, B, D and E Only

73. If $\vec{L}$ and $\vec{P}$ represent the angular momentum and linear momentum respectively of a particle of mass 'm' having position vector $\vec{r} = a(\hat{i}\cos\omega t + \hat{j}\sin\omega t)$. The direction of force is **[JEE-Mains-2025]**

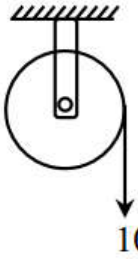
(A) Opposite to the direction of $\vec{r}$ (B) Opposite to the direction of $\vec{L}$

(C) Opposite to the direction of $\vec{P}$ (D) Opposite to the direction of $\vec{L} \times \vec{P}$

74. A rod of linear mass density 'λ' and length 'L' is bent to form a ring of radius 'R'. Moment of inertia of ring about any of its diameter is : **[JEE-Mains-2025]**

(A) $\frac{\lambda L^3}{16\pi^2}$ (B) $\frac{\lambda L^3}{12}$ (C) $\frac{\lambda L^3}{4\pi^2}$ (D) $\frac{\lambda L^3}{8\pi^2}$

75. A wheel of radius 0.2 m rotates freely about its center when a string that is wrapped over its rim is pulled by force of 10 N as shown in figure. The established torque produces an angular acceleration of 2 rad/s^2. Moment of inertia of the wheel is_____ kg m^2. (Acceleration due to gravity = 10 m/s^2) **[JEE-Mains-2025]**



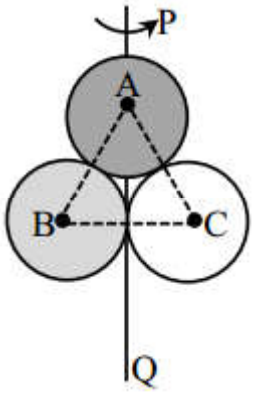
76. A solid sphere with uniform density and radius R is rotating initially with constant angular velocity (ω_1) about its diameter. After some time during the rotation its starts loosing mass at a uniform rate, with no change in its shape. The angular velocity of the sphere when its radius become R/2 is $x\omega_1$. The value of x is _____.

[JEE-Mains-2025]

77. A circular ring and a solid sphere having same radius roll down on an inclined plane from rest without slipping. The ratio of their velocities when reached at the bottom of the plane is $\sqrt{\frac{x}{5}}$ where x = _____

[JEE-Mains-2025].

78. A,B and C are disc, solid sphere and spherical shell respectively with same radii and masses. These masses are placed as shown in figure. **[JEE-Mains-2025]**

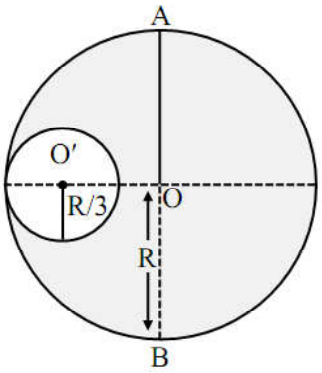


The moment of inertia of the given system about PQ is $\frac{x}{15}I$, where I is the moment of inertia of the disc about its diameter. The value of x is_____.

79. A thin solid disk of 1 kg is rotating along its diameter axis at the speed of 1800 rpm. By applying an external torque of 25 π Nm for 40s, the speed increases to 2100 rpm. The diameter of the disk is ____ m.

[JEE-Mains-2025]

80. M and R be the mass and radius of a disc. A small disc of radius R/3 is removed from the bigger disc as shown in figure. The moment of inertia of remaining part of bigger disc about an axis AB passing through the centre O and perpendicular to the plane of disc is $\frac{4}{x}MR^2$. The value of x is _____. **[JEE-Mains-2025]**



ANSWER KEY

EXERCISE-I

Only one option is correct

1.	B	**2.**	D	**3.**	D	**4.**	B	**5.**	A	**6.**	B	**7.**	D
8.	D	**9.**	A	**10.**	B	**11.**	B	**12.**	A	**13.**	B	**14.**	B
15.	C	**16.**	A	**17.**	D	**18.**	B	**19.**	C	**20.**	C	**21.**	B
22.	D	**23.**	B	**24.**	A	**25.**	D	**26.**	C	**27.**	A	**28.**	B
29.	D	**30.**	D	**31.**	C	**32.**	C	**33.**	B	**34.**	D	**35.**	A
36.	A	**37.**	D	**38.**	C	**39.**	D	**40.**	B	**41.**	B	**42.**	B
43.	B	**44.**	D	**45.**	B	**46.**	B						

One or more than one option may be correct

47. ACD **48.** ABC **49.** C **50.** BC **51.** BCD

52. (A) $\rightarrow$ R ; (B) $\rightarrow$ P ; (C) $\rightarrow$ Q ; (D) $\rightarrow$ S **53.** (A) $\rightarrow$ Q ; (B) $\rightarrow$ P ; (C) $\rightarrow$ S ; (D) $\rightarrow$ R

54. (A) $\rightarrow$ S ; (B) $\rightarrow$ S ; (C) $\rightarrow$ P ; (D) $\rightarrow$ P

EXERCISE-II

1. $\frac{ml^2}{12}$ **2.** $\frac{11ML^2}{24}$ **3.** At a distance $a/2\sqrt{3}$ from the centre down the plane.

4. 8 kg **5.** (i) $10/13 m/s^2$, (ii) $5000/26\pi$,(iii) 480/13N **6.** $6mv_0^2$

7. $M = 2m\left(\frac{2gh}{R^2\omega^2} - 1\right)$ **8.** gr/bh **9.** (a) $\frac{9g}{7}\downarrow$ (b) $\frac{4mg}{7}\uparrow$ **10.** 1/2 ma **11.** 4R

12. 10/3 m/s **13.** 5 N . m **14.** $\frac{\omega}{3}$ **15.** 120 **16.** 2

17. 2 m/s **18.** $16 m/s^2$ **19.** $v = \sqrt{\frac{27}{7}gR}$ **20.** $-\frac{1}{15}v_0, \frac{8v_0}{5\ell}$

EXERCISE-III

1.	2	**2.**	4	**3.**	A	**4.**	D	**5.**	D	**6.**	D	**7.**	ABC
8.	A	**9.**	B	**10.**	BC	**11.**	0.75	**12.**	A	**13.**	B	**14.**	A
15.	B	**16.**	ACD	**17.**	C	**18.**	BD	**19.**	ABD	**20.**	49.00	**21.**	120
22.	B	**23.**	0.52	**24.**	2.86	**25.**	176	**26.**	A	**27.**	ABCD	**28.**	30.00
29.	A	**30.**	C	**31.**	B	**32.**	D	**33.**	8.00	**34.**	250.00	**35.**	53.00
36.	7.00	**37.**	60	**38.**	24.00	**39.**	6.00	**40.**	16.00	**41.**	10.00	**42.**	4.00
43.	C	**44.**	7	**45.**	2	**46.**	13	**47.**	4	**48.**	100	**49.**	3
50.	18.00	**51.**	12.00	**52.**	B	**53.**	D	**54.**	C	**55.**	A	**56.**	B
57.	A	**58.**	B	**59.**	C	**60.**	A	**61.**	D	**62.**	C	**63.**	90
64.	16.00	**65.**	4.00	**66.**	2	**67.**	A	**68.**	C	**69.**	B	**70.**	A
71.	A	**72.**	C	**73.**	A	**74.**	D	**75.**	1.00	**76.**	32.00	**77.**	04.00
78.	199.00	**79.**	40.00	**80.**	9.00								